



one store (card_games) · one metric (ruling_intensity) · day granule
 rule same_valid_time_window over one anchor pair · pinned version $v = v1$
 num leg @ rulings.date Cov = [2004-10-04, 2021-02-05] hull over 87,769 rows
 den leg @ sets.releaseDate Cov = [1993-08-05, 2021-03-19] hull over 551 rows

(a) Bindable

CARD-Q2 · certificate ACCEPT

$T = 2016-11-15$ · $W_n = W_d = [11-01, 12-01]$

✓ $W \cap \text{Cov} \neq \emptyset$ ✓ svw (i)-(iv) hold ✓ (i) ok · (ii) $\mu_d = 796 > 0$

$\llbracket \mathbf{q} \rrbracket = \mu_n / \mu_d = 320 / 796 = 40.20\%$

(b) $\perp$ OOV

probe on the same store · no certificate

$T = 2021-06-15$ · $W_n = W_d = [06-01, 07-01]$

✗ $W \cap \text{Cov} = \emptyset$ - not reached - not reached

witness (a, W_T , π_V): the request window opens after both hulls close

(c) $\perp$ AM(ii)

probe on the same store · no certificate

$T = 2016-11-15$ · $W_n = [11-01, 12-01]$ $W_d = [10-01, 11-01]$ (imperative pair)

✓ $W \cap \text{Cov} \neq \emptyset$ ✗ svw (ii): $W_n \neq W_d$ - not reached

as asked $320/5 = 6400.00\%$ · same window $320/796 = 40.20\%$

(d) $\perp$ MC(ii)

CARD-Q7 · certificate ACCEPT

$T = 2017-02-15$ · $W_n = W_d = [02-01, 03-01]$

✓ $W \cap \text{Cov} \neq \emptyset$ ✓ svw (i)-(iv) hold ✗ (i) ok · (ii) $\mu_d = 0$

witness: denominator probe, $\mu_d = 0$ printings, against $\mu_n = 2,161$ rulings

(b) and (d) both face a zero-release month, and still split. 2017-02 lies inside the release anchor's hull (280 printings in Jan-17, 314 in Mar-17, none in Feb), so its empty denominator is a caliber failure; 2021-06 opens after both hulls close (2021-02-05 / 2021-03-19), so it never reaches the caliber guard. The discriminator is the anchor's hull — a property of its table.